

XBoing Screen State Machine: A Z Specification

Formal Model of the Flat Mode Machine and its Shared Resources

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1 Introduction

The XBoing screen machine is *flat*. `struct sdl2_state` (`src/sdl2_state.c`) holds a single `current` mode (`SDL2ST_*`), plus `previous` and a `saved` slot for dialogue push/pop. There is no enclosing “attract” super-state: `Intro`, `Instruct`, `Demo`, `Preview`, `Keys`, `KeysEdit` and `Highscore` are all peer top-level modes. The “attract loop” is a designated *subset* with a constrained transition order (`attract_cycle[]`), i.e. a constraint on a flat machine, not a nesting.

A second, orthogonal nesting axis does exist but is out of scope here: while `current` sits on an attract mode, that mode’s driving *system* runs its own animation phase (`intro_state_t`, `demo_state_t`, ...) held inside `ctx->intro` / `ctx->demo` / `ctx->keys`, not inside `sdl2_state`. Three systems are each shared by two modes (`intro_system` drives `Intro/Instruct`; `demo_system` drives `Demo/Preview`; `keys_system` drives `Keys/KeysEdit`).

Three resources are *shared* between the attract screens and real gameplay: the `game_active` flag, the required-block grid (`block_system_still_active`), and the high-score tables (`hs_global` / `hs_personal`, chosen by `highscore_request_type`). The observed defects all arise because a transition reads a shared resource that a prior transition left in the wrong state. This model names the safety invariants the handlers must preserve and shows which operations break them.

2 Free Types

The outer modes. Legacy `BallWait/Wait` (defined but never assigned) are omitted.

$$\begin{aligned} Mode ::= & mNone \mid mPresents \mid mIntro \mid mInstruct \\ & \mid mDemo \mid mPreview \mid mKeys \mid mKeysedit \\ & \mid mHighscore \mid mBonus \mid mGame \mid mPause \\ & \mid mDialogue \mid mEdit \end{aligned}$$

The shared inner systems (descriptive: the nesting axis noted above).

$$System ::= sysNone \mid sysIntro \mid sysDemo \mid sysKeys$$

Which high-score table the display is bound to (`highscore_request_type`).

$$Board ::= bGlobal \mid bPersonal$$

fuzz has no built-in boolean; `ZBOOL` avoids the `ProB/B` keyword clash with `BOOL`.

$$ZBOOL ::= ztrue \mid zfalse$$

3 Screen Sets and the Attract Cycle

The attract screens form the auto-cycle; `Presents` is not part of it.

$$\begin{array}{|l} attractModes : \mathbb{P} Mode \\ \hline attractModes = \{mIntro, mInstruct, mDemo, mPreview, \\ mKeys, mKeysedit, mHighscore\} \end{array}$$

Attract screens on which `Space` starts a game directly. `Intro` (goes to `Instruct`) and `Highscore` (conditional) are handled separately (`game_input.c:399--423`).

$$\begin{array}{|l} spaceStartModes : \mathbb{P} Mode \\ \hline spaceStartModes = \{mInstruct, mDemo, mPreview, mKeys, mKeysedit\} \end{array}$$

The cycle order used by the `C` key and each screen’s finish-timer auto-advance (`attract_cycle[]`, `game_callbacks.c:54--68`).

$attractNext : Mode \leftrightarrow Mode$
$attractNext = \{mIntro \mapsto mInstruct,$ $mInstruct \mapsto mDemo,$ $mDemo \mapsto mKeys,$ $mKeys \mapsto mKeysedit,$ $mKeysedit \mapsto mPreview,$ $mPreview \mapsto mHighscore,$ $mHighscore \mapsto mIntro\}$

The shared system driving each attract screen (two-to-one sharing).

$driver : Mode \leftrightarrow System$
$driver = \{mIntro \mapsto sysIntro,$ $mInstruct \mapsto sysIntro,$ $mDemo \mapsto sysDemo,$ $mPreview \mapsto sysDemo,$ $mKeys \mapsto sysKeys,$ $mKeysedit \mapsto sysKeys\}$

4 State

The flattened screen state. `blocksLoaded` abstracts `block_system_still_active` (required blocks present). `globalHasData` / `personalHasData` record whether each table held rows after the startup read (`game_init.c:327--335`); on an unprivileged install (no `/var/games`) `globalHasData` is `zfalse` while `personalHasData` may be `ztrue`. `boardChoice` is `highscore_request_type`.

$Screen$
$mode : Mode$ $gameActive : ZBOOL$ $blocksLoaded : ZBOOL$ $boardChoice : Board$ $globalHasData : ZBOOL$ $personalHasData : ZBOOL$

The machine boots into `Presents`: `ctx` is `calloc`'d so `gameActive` is `zfalse` (`game_init.c:231`); no level is loaded; both boards are read from disk (`game_init.c:327--335`) so the data flags reflect what was on disk — left unconstrained here because they are platform-dependent; and `boardChoice` defaults to `bGlobal` (`game_init.c:340`).

$Init$
$Screen'$
$mode' = mPresents$ $gameActive' = zfalse$ $blocksLoaded' = zfalse$ $boardChoice' = bGlobal$

5 Safety Invariants (the intended contract)

Stated as observer schemas, deliberately *not* folded into `Screen`: in Z the state predicate constrains every $\Delta Screen$, so folding an invariant in would silently *disable* the violating transitions rather than expose them (probcli then reports a spurious deadlock). Instead each contract is machine-checked by model-checking toward its *negation* as a probcli goal:

```
probcli spec.tex -model_check -goal "<negated invariant>"
```

A reachable state matching the negation is a counter-example proving the defect; “No counter example found. ALL states visited” after a fix proves the guard closes it.

A game may only run once its level’s required blocks are on the grid; otherwise the per-tick rule check reads an empty grid and declares the level complete.

<i>SafeGame</i>	_____
<i>Screen</i>	_____
$mode = mGame \Rightarrow blocksLoaded = ztrue$	

The high-score screen must not display an *empty* board when a populated one is available: if it is bound to an empty global board, the personal board must also be empty (nothing better to show).

<i>SafeHighscore</i>	_____
<i>Screen</i>	_____
$(mode = mHighscore$ $\quad \wedge boardChoice = bGlobal$ $\quad \wedge globalHasData = zfalse)$ $\Rightarrow personalHasData = zfalse$	

`game_active` may legitimately stay true on the game-over **Highscore** display (it gates score submission), but it must not propagate to the *other* attract screens; otherwise the **Highscore Space** routing misfires.

<i>SafeAttract</i>	_____
<i>Screen</i>	_____
$(mode \in attractModes \wedge mode \neq mHighscore)$ $\Rightarrow gameActive = zfalse$	

6 Operations (as coded)

6.1 Boot: Presents auto-advances to Intro

The machine boots into Presents; when the credits animation finishes, `presents_cb_on_finished` (`game_callbacks.c:596--600`) transitions unconditionally to Intro, entering the attract cycle. Presents is not part of `attract_cycle[]`.

<i>Boot</i>	_____
$\Delta Screen$	_____
$mode = mPresents$ $mode' = mIntro$ $gameActive' = gameActive$ $blocksLoaded' = blocksLoaded$ $boardChoice' = boardChoice$ $globalHasData' = globalHasData$ $personalHasData' = personalHasData$	

6.2 Attract auto-advance and the C key

Both the finish-timer (`highscore_cb_on_finished` and peers) and the C key advance the cycle. Crucially, this transition *preserves* `gameActive` — the auto-advance from **Highscore** does not clear it (`game_callbacks.c:750--759`). Entering **Highscore** binds the display to `boardChoice` but performs no disk read and no global-to-personal fallback on the attract path (`game_modes.c:1139--1142`).

AttractAdvance

Δ *Screen*

$mode \in \text{dom } \text{attractNext}$
 $mode' = \text{attractNext}(mode)$
 $(mode' = mIntro \Rightarrow gameActive' = zfalse)$
 $(mode' \neq mIntro \Rightarrow gameActive' = gameActive)$
 $blocksLoaded' = blocksLoaded$
 $(mode' = mHighscore \wedge gameActive = zfalse$
 $\quad \wedge globalHasData = zfalse \wedge personalHasData = ztrue$
 $\quad \Rightarrow boardChoice' = bPersonal)$
 $(\neg (mode' = mHighscore \wedge gameActive = zfalse$
 $\quad \wedge globalHasData = zfalse \wedge personalHasData = ztrue)$
 $\quad \Rightarrow boardChoice' = boardChoice)$
 $globalHasData' = globalHasData$
 $personalHasData' = personalHasData$

6.3 Starting a game — level load succeeds

`start_new_game` (`game_modes.c:90--166`) sets `gameActive` and, on a successful load, populates the grid via the `on_add_block` callback. This is the correct path.

StartNewGameOk

Δ *Screen*

$mode \in \text{spaceStartModes}$
 $mode' = mGame$
 $gameActive' = ztrue$
 $blocksLoaded' = ztrue$
 $boardChoice' = boardChoice$
 $globalHasData' = globalHasData$
 $personalHasData' = personalHasData$

6.4 Starting a game — level load fails silently

If the level file cannot be resolved or parsed (`game_modes.c:128--136`) `start_new_game` only warns to `stderr` and still enters `Game` with the grid left cleared.

StartNewGameFail

Δ *Screen*

$mode \in \text{spaceStartModes}$
 $mode' = mIntro$
 $gameActive' = zfalse$
 $blocksLoaded' = zfalse$
 $boardChoice' = boardChoice$
 $globalHasData' = globalHasData$
 $personalHasData' = personalHasData$

6.5 Space on the title

IntroSpace

Δ Screen

```
mode = mIntro
mode' = mInstruct
gameActive' = gameActive
blocksLoaded' = blocksLoaded
boardChoice' = boardChoice
globalHasData' = globalHasData
personalHasData' = personalHasData
```

6.6 Space on the high-score screen

The **Space** handler routes on `gameActive` (`game_input.c:411--423`): a live game-over returns to **Intro**; otherwise it starts a game. On the return to **Intro**, `gameActive` is cleared by `mode_intro_enter` — the single clear site every attract exit from **Highscore** lands on (ADR-055) — as the postcondition below records.

HighscoreSpaceReturn

Δ Screen

```
mode = mHighscore
gameActive = ztrue
mode' = mIntro
gameActive' = zfalse
blocksLoaded' = blocksLoaded
boardChoice' = boardChoice
globalHasData' = globalHasData
personalHasData' = personalHasData
```

HighscoreSpaceStart

Δ Screen

```
mode = mHighscore
gameActive = zfalse
mode' = mGame
gameActive' = ztrue
blocksLoaded' = ztrue
boardChoice' = boardChoice
globalHasData' = globalHasData
personalHasData' = personalHasData
```

6.7 Game over

`game_rules_ball_died` (`game_rules.c:294--296`) leaves `gameActive` true and shows **Highscore**.

GameOver

Δ Screen

```
mode = mGame
mode' = mHighscore
gameActive' = ztrue
blocksLoaded' = blocksLoaded
(globalHasData = zfalse  $\wedge$  personalHasData = ztrue
 $\Rightarrow$  boardChoice' = bPersonal)
( $\neg$  (globalHasData = zfalse  $\wedge$  personalHasData = ztrue)
 $\Rightarrow$  boardChoice' = bGlobal)
globalHasData' = globalHasData
personalHasData' = personalHasData
```

6.8 The per-tick rule check

Every Game tick, `game_rules_check` (`game_rules.c:330--340`) sends the machine to Bonus the instant no required blocks remain. It is *unconditional* — no guard that a level was actually loaded.

<i>GameRulesToBonus</i> $\Delta Screen$ <i>mode</i> = <i>mGame</i> <i>blocksLoaded</i> = <i>zfalse</i> <i>mode'</i> = <i>mBonus</i> <i>gameActive'</i> = <i>gameActive</i> <i>blocksLoaded'</i> = <i>blocksLoaded</i> <i>boardChoice'</i> = <i>boardChoice</i> <i>globalHasData'</i> = <i>globalHasData</i> <i>personalHasData'</i> = <i>personalHasData</i>

6.9 Bonus completes: next level loads

On bonus completion `bonus_cb_on_finished` advances to the next level (`game_rules_next_level`) and re-enters Game with the new level's blocks loaded (`game_callbacks.c:671--678`). Modelled so Bonus is not a terminal (deadlocking) mode.

<i>BonusToGame</i> $\Delta Screen$ <i>mode</i> = <i>mBonus</i> <i>mode'</i> = <i>mGame</i> <i>blocksLoaded'</i> = <i>ztrue</i> <i>gameActive'</i> = <i>gameActive</i> <i>boardChoice'</i> = <i>boardChoice</i> <i>globalHasData'</i> = <i>globalHasData</i> <i>personalHasData'</i> = <i>personalHasData</i>
--

7 Conjectures: where the invariants break

Each defect is a reachable state that violates one safety schema. A model checker (`probcli`) explores `Init` plus the operations and reports the trace; the prose states the trace and the missing guard.

Defect 2 — empty high-score board in the attract cycle (PROVEN, FIXED)

The boards *are* read at startup, so “never loaded” is not the cause. On an unprivileged install `globalHasData = zfalse` and `personalHasData = ztrue`, and `boardChoice = bGlobal` from `Init`. Goal-directed model checking of the negated `SafeHighscore` reached `mode = mHighscore`, `gameActive = zfalse` still bound to the empty global board by pure attract-cycling (*Init* ; *Boot* ; *AttractAdvance**), the machine-checked counter-example for the reported symptom.

Fix. `mode_highscore_enter` (`src/game_modes.c`) now falls back to `bPersonal` on the attract path (`!game_active`) when the global board is empty (`highscore_io_count(&hs_global) == 0`) and the personal board has scores. The game-over path keeps its own submit-based fallback (`game_modes.c:1134`). Both entry paths are modelled here in `AttractAdvance` and `GameOver`; re-running the goal now reports *No counter example found. ALL states visited* (138 states), so the fixed operations cannot reach the violating state. Regression: `test_highscore_attract_falls_back_to_personal` (`tests/test_integration_modes.c`) plus `highscore_io_count` unit tests.

Defect 3 — first Space advances the loop (PROVEN, FIXED)

In a *pure* attract cycle `gameActive` is always `zfalse` (set only by `StartNewGame*`, cleared only by the `Highscore Space` handler), so `IntroSpace` (`Intro` → `Instruct`) is the *intended* PR #172 behavior, not a bug. The genuine violation requires a prior game over: goal-directed model checking of the negated `SafeAttract`

found the 7-step trace ... ; *GameOver* ; *AttractAdvance* carrying `gameActive = ztrue` out of *Highscore* into *Intro* (*SafeAttract* violated at *Intro*); the next time the cycle reaches *Highscore*, *HighscoreSpaceReturn* fires and the first *Space* goes to *Intro* instead of starting a game.

Fix. Every attract advance out of *Highscore* — the finish-timer (`highscore_cb_on_finished`), the C cycle key (`game_input.c`), and the *Space* return — lands on *Intro*, since `attract_next(HIGHSCORE) == INTRO`. A single clear in `mode_intro_enter` (`src/game_modes.c`) therefore closes all three leak paths at one site; modelled here as *AttractAdvance* setting `gameActive' = zfalse` whenever `mode' = mIntro`. Re-running the goal now reports *No counter example found*. *ALL states visited* (90 states), and no non-*Highscore* attract mode is reachable with `gameActive = ztrue`. Regression: `test_highscore_autoadvance_clears_game_active` (timer path) and `test_highscore_c_cycle_clears_game_active` (C key). See ADR-055.

Defect 1 — Space lands in Bonus with an empty grid (PROVEN, FIXED)

StartNewGameOk establishes `blocksLoaded`, so the model exonerates the *happy-path* *Space*-to-game edge. Goal-directed model checking of the negated *SafeGame* reached `mode = mGame`, `blocksLoaded = zfalse` via *StartNewGameFail* — a silent level-load failure (`start_new_game` ignored the load result and entered *Game* with the grid cleared); the next *GameRulesToBonus* tick then dropped to *Bonus*.

Fix. `start_new_game` (`src/game_modes.c`) now checks `level_system_load_file` and returns failure without setting `game_active`; `mode_game_enter` refuses *Game* and returns to *Intro*. This preserves the original's guarantee — `original/file.c:142--146` (*SetupStage* exits via *ShutDown* on a *ReadNextLevel* failure) never let gameplay run on an unloaded grid — modernising only the mechanism (mode-refusal, not process exit). Modelled as *StartNewGameFail* now targeting `mIntro` with `gameActive = zfalse`; the goal is unreachable (76 states, *No counter example found*). Regression: `test_game_aborts_on_level_load_failure`. See ADR-056.

8 Verification summary

Each invariant was machine-checked by model-checking toward its negation as a `probcli` goal. In the as-coded model all three goals were reachable (counter-example traces above); after the fixes all three are unreachable in one exhaustive pass:

- *SafeHighscore* — attract *Highscore* falls back to the populated personal board when the global board is empty (`mode_highscore_enter`; ADR-054).
- *SafeAttract* — `game_active` is cleared on entering the title, so it cannot leak into the attract screens (`mode_intro_enter`; ADR-055).
- *SafeGame* — a failed level load refuses *Game* and returns to the title instead of entering with an empty grid (`start_new_game` / `mode_game_enter`; ADR-056).

```
probcli spec.tex -model_check -goal "<negated invariant>"
=> No counter example found. ALL states visited.
```

The mode suite previously force-transitioned to each mode and ticked for *no-crash* only, asserting nothing about the shared resources. Each invariant is now both a re-proven model postcondition and a red→green regression test, named in its per-defect section above.